



HADROS User Manual

High-energy Astrophysical DIS Radiation and Observation Simulator

HADROS

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1 Overview

HADROS combines Kerr ray tracing, semi-analytic astrophysical backgrounds, UHE neutrino DIS opacity, phenomenological source and spectral prescriptions, diagnostic MeV neutrino post-processing, ParaView export, and a static dashboard for organizing plots.

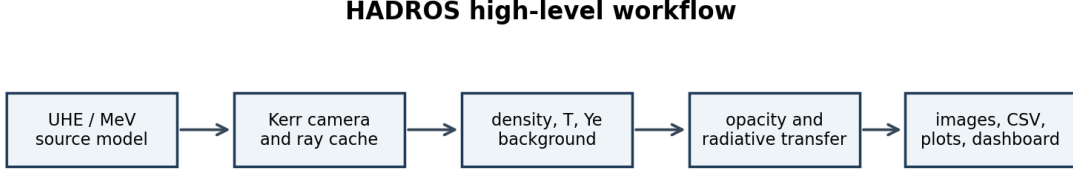


Figure 1: High-level HADROS workflow.

At a high level the pipeline is

source $\rightarrow$ Kerr rays $\rightarrow$ background $\rightarrow$ opacity/transfer $\rightarrow$ images and diagnostics.

HADROS does not perform hydrodynamic evolution. It is not full neutrino radiation hydrodynamics, not a calibrated collapsar simulation, and not a complete PDF-uncertainty package. The current backgrounds are semi-analytic density morphologies, the UHE sources are phenomenological prescriptions, and the MeV luminosities are diagnostic proxies.

2 Physical Theories

2.1 Kerr Ray Tracing

The spacetime around a rotating black hole is modeled with the Kerr metric. The main physical parameters are the black-hole mass `MBH_MSUN` and the dimensionless spin `ASPIN`. The observer is placed at a large radius `CAM_R_OBS_RG` and inclination `CAM_THETA_DEG`. The camera launches rays backward through the Kerr geometry and stores geodesics in a cache.

For neutrinos in the UHE range, the rest mass is negligible compared with the energy, so the propagation is approximated by null geodesics:

$$ds^2 = 0.$$

The ray path is therefore controlled by the Kerr geometry. The energy dependence in UHE images mostly enters through opacity, $\sigma_{\nu N}(E)$, not through chromatic lensing.

2.2 Semi-analytic Density Backgrounds

The matter field is a controlled semi-analytic background, not a hydrodynamic solution. Available density profiles include `gaussian`, `powerlaw`, funnel variants, envelope variants, and `collapsar_ndaf_like`.

Every profile applies an explicit density floor:

$$\rho = \max(\rho_{\text{raw}}, \rho_{\text{floor}}).$$

Regime	Purpose	Interpretation and limitation
<code>fiducial_uhe_default</code>	UHE baseline	Low-density transparent test background; not a collapsar disk.
<code>fiducial_mev_density</code>	MeV baseline	Around 10^9 – 10^{10} g cm $^{-3}$ in audits; weakly cooled torus with expected low luminosity.
<code>collapsar_ndaf_like</code>	high-density preset	Significant regions near 10^{10} – 10^{12} g cm $^{-3}$; semi-analytic NDAF/collapsar-like regime, not hydrodynamics.

2.3 UHE Neutrino DIS Opacity

The UHE optical depth is computed from the baryon column along a ray:

$$\tau_{\text{DIS}}(E) = \int n_b(s) \sigma_{\nu N}(E) ds, \quad P_{\text{surv}}(E) = \exp[-\tau_{\text{DIS}}(E)].$$

HADROS currently compares `GBW`, `IIM`, `CTW_reference`, and `literature_powerlaw_scale`. `CTW_reference` is the central νN charged-current table from Connolly, Thorne & Waters, Phys. Rev. D 83, 113009, Table I. It does not include neutral-current interactions, antineutrino tables, regeneration, or PDF uncertainty bands.

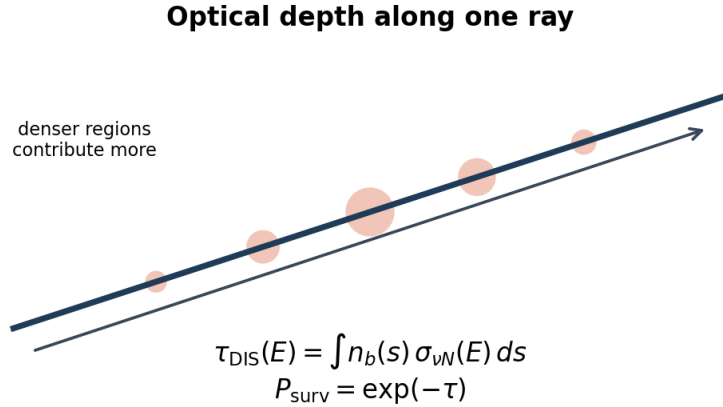


Figure 2: Optical depth and survival probability along a ray.

2.4 UHE Source Morphologies

The UHE emissivity is separated into a source morphology and a spectral model. Implemented source morphologies are `inner_ring`, `funnel_wall`, `jet_base`, `shock_layer`, and `density_weighted`. These are phenomenological source morphologies, not first-principles particle-acceleration simulations.

2.5 UHE Spectral Models

The spectral interface currently supports `monochromatic`, `powerlaw`, and `powerlaw_cutoff`. The power-law spectrum is

$$\frac{dN}{dE} \propto E^{-\gamma},$$

and the cutoff model is

$$\frac{dN}{dE} \propto E^{-\gamma} \exp(-E/E_{\text{cut}}).$$

Energy-band composite images use false colors: blue for low-energy UHE, green for intermediate-energy UHE, and red for high-energy UHE.

2.6 Optical-depth Surfaces

HADROS can extract diagnostic opacity surfaces such as $\tau = 2/3$, $\tau = 1$, and $\tau = 3$. These surfaces are properties of the medium, geometry, energy, and DIS cross section. They do not depend on the UHE source morphology. The current extraction is axisymmetric and returns $r_\tau(\theta)$.

2.7 Thermal MeV Neutrino Module

The MeV module is separate from UHE/DIS. It is a diagnostic local emissivity and radiative-transfer model, not a calibrated absolute luminosity prediction. Implemented channels include URCA-like processes,

$$e^- + p \rightarrow n + \nu_e, \quad e^+ + n \rightarrow p + \bar{\nu}_e,$$

pair annihilation,

$$e^- + e^+ \rightarrow \nu + \bar{\nu},$$

and nucleon-nucleon bremsstrahlung,

$$N + N \rightarrow N + N + \nu + \bar{\nu}.$$

The local transfer equation is

$$\frac{dI}{ds} = j - \alpha I.$$

CPU MeV is the canonical physical reference implementation. CUDA MeV remains legacy/toy until it is ported and validated.

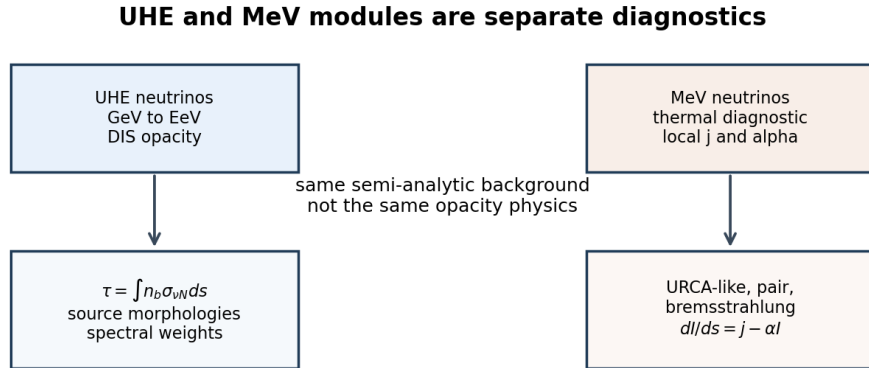


Figure 3: UHE and MeV modules are separate diagnostics.

2.8 MeV Diagnostic Upgrades

The MeV workflow includes temperature profiles, electron-fraction profiles, Fermi-Dirac-like spectral integration, MeV neutrinosphere extraction at $\tau_{\text{MeV}} = 2/3$, electron-degeneracy diagnostics, opacity-component decomposition, and luminosity-proxy audits.

2.9 ParaView and 3D Visualization

The ParaView export samples the current axisymmetric semi-analytic model onto a 3D Cartesian grid. It is not a fully 3D hydrodynamical simulation and does not export a fake local optical-depth field.

3 Main Makefile Commands

3.1 Safe Build and Validation

```
make
make build NTHREADS=2
make validate_small NTHREADS=2
make validate_small_cuda NTHREADS=2 \
    NVCC=/home/rafael/micromamba/envs/dis/bin/nvcc \
    PYTHON=/home/rafael/micromamba/envs/dis/bin/python
make validate_medium_cuda NTHREADS=2 \
    NVCC=/home/rafael/micromamba/envs/dis/bin/nvcc \
    PYTHON=/home/rafael/micromamba/envs/dis/bin/python
```

3.2 Running Images

```
make image-from-cache ENU=1e11 DENSITY_PROFILE=gaussian NTHREADS=2
make image-from-cache SPECTRAL_MODEL=powerlaw \
    SPECTRAL_GAMMA=2.0 SPECTRAL_E_MIN_GEV=1e5 \
    SPECTRAL_E_MAX_GEV=1e12 SPECTRAL_N_BINS=8 NTHREADS=2
make plot_multiband_image PYTHON=python3
make image-from-cache MEV_MODEL=physical MEV_FLAVOR=anti_nu_e NTHREADS=2
make mev_multiband_image NTHREADS=2 PYTHON=python3
```

3.3 Robustness, DIS, Opacity, and MeV Diagnostics

```
make robustness_scans NTHREADS=2 PYTHON=python3
make validate_dis_models NTHREADS=2 PYTHON=python3
make opacity_surfaces TAU_SURFACE_VALUE=1.0 PYTHON=python3
make validate_mev_physics PYTHON=python3
make mev_neutrinosphere PYTHON=python3
make mev_luminosity PYTHON=python3
make audit_mev_luminosity PYTHON=python3
make audit_torus_regime PYTHON=python3
make audit_collapsar_ndaf_like PYTHON=python3
```

3.4 ParaView and Dashboard

```
make paraview_fields \
    DENSITY_PROFILE=powerlaw_funnel_envelope \
    SOURCE_MODEL=funnel_wall \
    PARAVIEW_NX=64 PARAVIEW_NY=64 PARAVIEW_NZ=64 \
    PARAVIEW_BOX_RG=80 NTHREADS=4

make dashboard PYTHON=python3
```

Open `dashboard/index.html` in a browser. The dashboard indexes existing plots and output products; it does not run simulations.

3.5 Thread Control

Use explicit limits:

```
make -j2 build
make validate_small NTHREADS=2
make run_production NTHREADS=4
```

The default `make` target prints help and does not launch a production simulation.

4 Outputs and Directory Map

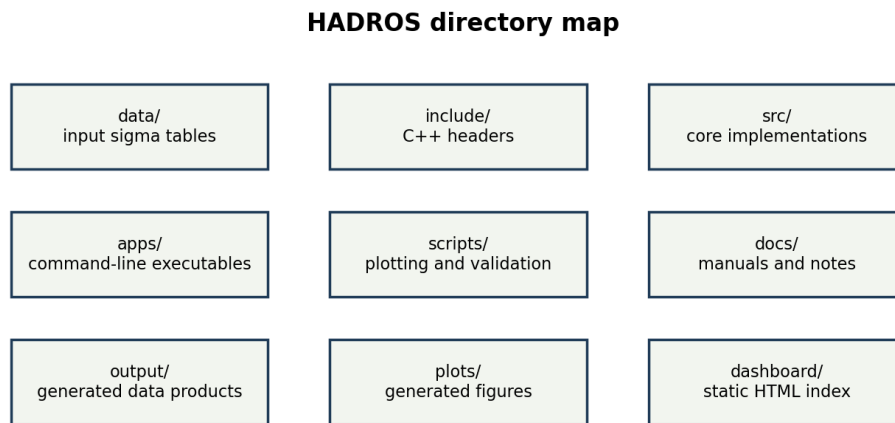


Figure 4: Main HADROS directories.

- `data/`: input data tables, especially DIS sigma tables.
- `include/`: C++ headers.
- `src/`: core implementations.
- `apps/`: command-line applications.
- `scripts/`: plotting, validation, and audit scripts.
- `docs/`: manuals and scientific notes.
- `plots/`: generated figures.
- `output/`: generated tables, images, ray caches, and reports.
- `dashboard/`: static HTML dashboard.

5 Program Workflow

The internal workflow is:

1. choose a physical setup;
2. generate or load a Kerr ray cache;
3. evaluate the local background along each ray;
4. compute local UHE and/or MeV emissivity;
5. compute opacity;
6. integrate radiative transfer along rays;
7. write data tables;
8. generate plots;
9. index products in the dashboard.

6 Recommended Workflows for Students

Beginner dashboard:

```
make dashboard PYTHON=python3
```

Beginner validation:

```
make validate_small NTHREADS=2 PYTHON=python3
```

Intermediate DIS comparison:

```
make validate_dis_models NTHREADS=2 PYTHON=python3
```

Intermediate MeV diagnostics:

```
make validate_mev_physics PYTHON=python3
```

Advanced collapsar/NDAF-like comparison:

```
make audit_collapsar_ndaf_like PYTHON=python3
```

Advanced ParaView export:

```
make paraview_fields DENSITY_PROFILE=collapsar_ndaf_like NTHREADS=4
```

7 Scientific Limitations

- Backgrounds are semi-analytic, not hydrodynamic evolutions.
- HADROS does not solve full neutrino radiation hydrodynamics.
- MeV luminosities are proxy/diagnostic quantities.
- CUDA MeV remains legacy/toy and is not equivalent to CPU physical MeV.
- CTW_reference is central charged-current νN only.
- UHE neutral-current interactions and regeneration are not yet included.
- There is no true 3D hydrodynamic background yet.

8 Troubleshooting

If `make` seems to run too long, stop the process and use a validation target such as `validate_small`. If CPU usage is too high, pass `NTHREADS=2` or `NTHREADS=4`, and use limited make parallelism such as `make -j2 build`. If CUDA is not found, check `which nvcc`, `nvcc -version`, and `nvidia-smi`. If plots are missing, regenerate the relevant plot target and rebuild the dashboard. If stale caches are suspected after a crash, see `README_RECOVERY_20260609.md`.

9 Glossary

DIS Deep inelastic scattering, the high-energy neutrino-nucleon interaction used for UHE opacity.

UHE Ultra-high energy, here referring to GeV-to-EeV scale neutrinos.

MeV Thermal neutrino energy scale used in the diagnostic module.

Optical depth Dimensionless attenuation measure, τ .

Survival probability $P_{\text{surv}} = \exp(-\tau)$.

Neutrinosphere Diagnostic surface where optical depth reaches a chosen value, often $\tau = 2/3$.

Kerr metric Spacetime geometry around a rotating black hole.

NDAF Neutrino-dominated accretion flow.

Collapsar Massive-star collapse scenario that can form a black hole and dense accretion environment.

CTW Connolly, Thorne & Waters reference cross-section table.

GBW Golec-Biernat-Wusthoff saturation-inspired DIS model.

IIM Iancu-Itakura-Munier saturation-inspired DIS model.